

Midterm Exam Grading Rubric

IEDA 1250: Optimization for Decision Making (Summer, 2026)

Question 1: Linear Programming – Graphical Method (10 Points)

- Correctly identifies that the feasible region is a diamond bounded by $0 \leq x_1 + 2x_2 \leq 6$ and $-2 \leq 2x_1 - x_2 \leq 4$. **(2 Points)**
- Correctly finds the four vertices of the feasible region:

$$\left(-\frac{4}{5}, \frac{2}{5}\right), \left(\frac{8}{5}, -\frac{4}{5}\right), \left(\frac{2}{5}, \frac{14}{5}\right), \left(\frac{14}{5}, \frac{8}{5}\right).$$

(4 Points)

- Correctly identifies the optimal solution and optimal objective value:

$$x_1^* = \frac{14}{5}, \quad x_2^* = \frac{8}{5}, \quad z^* = \frac{18}{5}.$$

(4 Points)

Question 2: Linear Programming – Complementary Slackness (20 Points)

- Correctly writes or uses the dual constraints corresponding to the primal LP, taking into account that x_3, x_4, x_5 are unrestricted in sign. **(4 Points)**
- Correctly applies complementary slackness to the primal constraints and obtains

$$2x_1 + x_2 + x_3 + x_4 = 3.$$

(4 Points)

- Correctly applies complementary slackness to the dual constraints for x_1 and x_2 , and concludes that

$$x_1 = 0, \quad x_2 = 0.$$

(5 Points)

- Correctly substitutes $x_1 = 0$ and $x_2 = 0$ into the primal constraints and forms the reduced system:

$$\begin{cases} x_3 + x_4 = 3, \\ 2x_3 + x_5 = 7, \\ x_3 - x_4 + 2x_5 = 7. \end{cases}$$

(4 Points)

- Correctly solves the reduced system and obtains the optimal primal solution:

$$x_1^* = 0, \quad x_2^* = 0, \quad x_3^* = 2, \quad x_4^* = 1, \quad x_5^* = 3.$$

the optimal objective value:

$$z^* = 6.$$

(3 Points)